

ZTA Solution Guide

The reference answer for every use case — pain-point responses, requirement responses, and expected products (from the official Solution Guide). Use it alongside the **ZTA Grading Guide** (rubric, levels & scoring). **Do not show participants.**

This document holds the **answers**. For *how to grade* — the Associate/Professional/Expert rubric, the expected answer at each level, and the score & feedback sheets — see the **ZTA Grading Guide**.

Use Case 1 — Managed Environment — Segmentation

Required

100 pts

Segment campus & branch wired/wireless clients into groups (Employees, IOT, deskagents, Guest) and control what each can reach. Pains: VLAN/subnet complexity, labour-intensive profiling, failing PCI, PCs behind on AV/OS, no budget for more firewalls.

Reference answer — addressing pain points

CUSTOMER PAIN POINT	EXPECTED CISCO RESPONSE
Prior segmentation failed. Using VLANs/IP subnets is complex and tedious.	Implement SGTs instead of VLAN/IP subnets.
Profiling is labor intensive; staff can't keep up with new devices.	ISE cloud AI profiling.
Failing PCI audits.	Apply SGT for segmentation and insert FW services where L7 security is required.
Corporate PCs not running latest AntiVirus and OS updates.	Use ISE Posture and remediate out-of-date PCs.
No budget for buying more firewalls to do segmentation.	Use SGTs to enforce segmentation within the wired/wireless network.

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	EXPECTED CISCO RESPONSE
Employees / IOT / deskagents (Mac-minis) / Guest groups.	Assign an SGT for each group. Authenticate employees via 802.1X or WebAuth; profile Mac-minis (or BYOD/MAB); profile IOT via MAB or manual SGT assignment.
IOT restricted from other groups but needs data center and Internet.	Configure the SGT Matrix to allow IOT → Data Center SGT and Internet only; deny all other destination SGTs.
Guests should have Internet access only.	Configure the SGT Matrix to allow SGT=Guest → SGT=Internet only.
Employee and IOT require controlled internet and data center application access.	Configure the SGT Matrix per source & destination SGTs.
Deskside Mac-Minis access on-prem data center only.	Configure the SGT Matrix: SGT=MacMini permit to SGT=on-prem-DC only.

Reference answer — expected products

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
ISE	Authenticate and profile every user and device (Web, 802.1X, MAB).	Visibility and scalable.

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
ISE-SGT	Assign an SGT to each group (Employee, IOT, Desktop-Mini, Internet).	Cisco innovation; dynamic policy and topology-independent; end-to-end segmentation into multicloud data centers (on-prem and cloud providers).
ISE SGT ACL policy	Granular access policy.	SGACL more efficient than IP ACLs.
ISE pxGrid	pxGrid / Rapid Threat Containment; shares context with ecosystem devices.	Cisco innovation, RFC8600; dynamically quarantines discovered threat devices; integrates with the Cisco portfolio (ACI, FW, SNA, SD-WAN, CSW, SA, switches, routers).
Catalyst SD-WAN	pxGrid integration for unified security policies via C3 / Security Cloud Control.	Consistent SGT policy across the WAN fabric.
FMC / SCC / FTD	Threat detection/protection via NGIPS, malware protection and content filtering.	Integrates with ISE via pxGrid for Rapid Threat Containment.
FMC / SCC / FTD — source/dest SGT enforcement	Dynamically blocks malicious hosts via SGT matching.	Cisco innovation; dynamic policy enforcement.
Cisco Secure Workload — App Dependency Mapping	Auto-profiles traffic flows and builds application-segmentation policy; ingests NetFlow from FTD.	Auto policy creation, validation and enforcement.
Cisco Secure Workload — pxGrid/ISE integration	pxGrid / ISE connector.	Builds granular segmentation policy based on identity and posture from ISE/pxGrid.

Use Case 2 — Remote Workers — Zero Trust Access

Required

100 pts

SASE for remote workers & branches. Requirements: ZTNA for modern apps (VPN only for legacy), block gambling/gaming, contractors clientless RDP, minimal sign-ons, SASE branches. Pains: over-permissive VPN, manual daily VPN, too many sign-ons, kill passwords, Okta too costly, reduce branch cost.

Reference answer — addressing pain points

CUSTOMER PAIN POINT	EXPECTED CISCO RESPONSE
Full client VPN over-permissive; not Zero Trust.	Use Zero Trust Access from Secure Client to connect to Secure Access.
Manual daily VPN; low satisfaction.	Use the Zero Trust Access module to automatically deliver a per-application VPN tunnel to corporate applications.
Too many sign-ons per application.	Use Duo Passport for SSO.
Want to eliminate password logins / stolen credentials.	Use Duo advanced passwordless.
Existing Okta IdP too expensive.	Use Duo Directory.
CIO wants to modernize branches with SASE.	Integrate SD-WAN into the Secure Access cloud with Direct Internet Access.

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	EXPECTED CISCO RESPONSE
Full client VPN for legacy apps; ZTNA for modern apps.	Use the ZT module in Secure Client and create private resources with ZT-module-only access; keep full VPN client access for other legacy corporate apps.
Controlled internet access (block gambling and gaming).	Use Secure Web Gateway (SWG) to block the gambling and gaming categories for corporate employees.
Contractors: clientless RDP access only.	Configure selected private resources to be accessible via browser (TCP) only for contractor groups.
Minimal application sign-ons.	Use Duo Passport or SSO.
Modernize branches with SASE.	Cisco Secure Access integrates with Catalyst SD-WAN to deliver Layer-7 services from the cloud.

Reference answer — expected products

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
Secure Access + Secure Client — ZTA module	The ZT Access module delivers per-app VPN access to private resources.	Ease of use; no full VPN required; meets Zero Trust least-privilege access.

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
Secure Client VPN module (Full Tunnel)	VPN module provides access to legacy applications.	Mature VPN technology; assigned an SGT for segmentation.
ZTA module (Secure Client)	Intercepts all ports/protocols from the client and redirects to Secure Access for inspection.	Control internet-bound traffic with SWG policy; block gambling and gaming.
Secure Access Private Resources proxy	Proxy for private resources; publishes apps for VPN client, ZTA module, or browser-only (guests, contractors — SSH/RDP).	Easy provisioning of private apps for various user types.
Duo	Duo Passport with SSO.	OS-level SSO login is unique — one login experience.
Secure Access VPNaaS	VPNaaS over on-prem VPN hardware at HQ.	VPN traffic can map to an SGT for segmentation — unique to Cisco.
Secure Access Unified Policy	Unified policy for public and private resources; integrated with SCC/CCC.	Unique to Cisco; ease of use.
Secure Access + SD-WAN	SASE integration with SD-WAN via IPsec tunnels.	Ease of deployment from SD-WAN to Secure Access.

Use Case 3 — AI Access and Agent Controls

Required

100 pts

Report and guardrail AI usage from Use Case 2. Requirements: fine-grained tool control, corporate ChatGPT tenant only, control Claude Desktop agents, govern MCP servers (Claude Code). Pains: shadow AI, unknown/over-privileged MCP servers, personal ChatGPT credentials, risky model downloads.

Reference answer — addressing pain points

CUSTOMER PAIN POINT	EXPECTED CISCO RESPONSE
Shadow AI usage is an issue today.	Use Secure Access's built-in AI Access feature to begin controlling corporate AI usage.
Over-privileged / unknown MCP servers in private and public DCs.	Use Secure Access's AI Gateway to register and manage allowed MCP servers.
Users using personal credentials for non-enterprise ChatGPT login.	Use Secure Access's tenant controls for AI.
Downloading of risky AI models.	Secure Access semantic inspection identifies and blocks malicious AI models.

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	EXPECTED CISCO RESPONSE
Fine-grained control; users not granted tool rights equally.	Duo / Secure Access MCP integration provides fine-grained AI-tools control along with authentication.
Only the organization's ChatGPT tenant.	Secure Access tenant control enforces the corporate ChatGPT tenant.
Controls for Claude Desktop agents on employee systems.	Duo IdP provides the agent registration process (DCR) to manage and control what registered desktop agents can do.
MCP servers integrated to local agents (Claude Code).	Register and govern MCP servers through the Secure Access AI Gateway with Duo authorization.

Reference answer — expected products

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
AI Access within Secure Access	AI control with guardrails; tenant control enforces the corporate ChatGPT tenant and semantic inspection blocks risky models.	Cisco integrates the DLP of Secure Access with Cisco AI Defense — monitors AI usage, enforces guardrails, and protects data across endpoints, cloud and web.
AI Gateway within Secure Access	AI Gateway registers and controls the allowed MCP servers.	Agents only reach approved tools.
Secure Client with agentic client discovery	Discovers and controls agents on employee systems (e.g. Claude Desktop).	Extends control to local desktop agents.

PRODUCT / FEATURE

HOW IT WORKS

WHY / DIFFERENTIATOR

Duo IdP

Fine-grained authorization and
agentic registration (DCR); OAuth
for MCP services to authorize
agent activities.

Easiest integration with the Secure Access AI
Gateway, plus flexibility to integrate 3rd-party
IdPs.

Use Case 4 — Merger & Acquisition — Extend Segmentation

Bonus

100 pts

Integrate a ~100-staff acquisition fast. Remote site has Cisco FTD (Internet edge + AI-cluster segmenting); acquired co uses Okta. Requirements: extend TrustSec segmentation, only selected users reach Site 9951. Pains: how to extend segmentation, don't want to onboard 100 users, minimize auth disruption, breached/passwords.

Reference answer — addressing pain points

CUSTOMER PAIN POINT	EXPECTED CISCO RESPONSE
Unsure how to extend segmentation to the acquired company.	Use SGT capability in the remote FTD and enable inline tagging to extend SGT tags from HQ to the remote company.
Not ready to add 100 acquired staff to the corporate directory — quick option?	Use Duo Directory to host the 100 acquired staff; Duo can import users from Okta via SCIM.
Minimize disruptive authentication during transition.	Use Duo SSO routing rules for a seamless experience; apps protected by Okta can still authenticate with Okta until the customer is ready to migrate off Okta.
Acquired company breached; still using passwords.	Use Duo Passwordless.

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	EXPECTED CISCO RESPONSE
Extend segmentation to the acquired company using TrustSec.	Use SGT capability in the remote FTD and enable inline tagging to extend SGT tags from HQ to the remote company.
Only selected branch users → restricted resource (Site 9951).	Use source & destination SGTs in the remote FTD access-control policy.

Reference answer — expected products

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
FMC / FTD	FTD builds a site-to-site VPN tunnel from the remote company to the HQ FTD; FTD supports inline SGT per interface.	Cisco SGT inline tagging is a differentiator over others.
Duo — IdP (Directory)	Duo Directory; use SCIM to import users from Okta easily.	Duo ease of use and integration bring overall value.
Duo — IdP (SSO routing)	Duo SSO routing rule maintains a consistent user-auth experience from old Okta to new.	Duo integrates smoothly with other IdPs.
Duo Passwordless	Immediately hardens the breached, password-based acquired users.	Phishing-resistant.

PRODUCT / FEATURE

HOW IT WORKS

WHY / DIFFERENTIATOR

SCC

SCC configures SGT policy into SD-WAN — match source SGT (branch) and destination SGT (acquisition FTD).

SCC provides unified SGT policy management.

Use Case 5 — Zero Trust across Multicloud Data Center

Bonus

100 pts

Application sprawl across on-prem/AWS/Azure; discover apps (incl. Kubernetes), minimize blast radius, app run-time protection. Requirements: limited budget/native features/virtual appliances, end-to-end segmentation, minimize complexity. Pains: no app visibility, no segmentation plan, K8s latency/visibility, can't patch IoT, too many disparate consoles.

Note: the source Solution Guide's UC5 pain-response table repeats UC4's answers by mistake; corrected here from the official UC5 requirements-response and product tables.

Reference answer — addressing pain points

CUSTOMER PAIN POINT	EXPECTED CISCO RESPONSE
Minimal visibility of applications across multicloud data centers.	Use Cisco Secure Workload App Dependency Mapping to discover apps and flows across on-prem, AWS and Azure.
Lacking segmentation strategy and plan.	Secure Workload auto-baseline policy creation — discover, validate, then enforce microsegmentation.
Wide Kubernetes deployment with latency and visibility issues.	Isovalent (Cilium CNI, eBPF) for Kubernetes observability and protection; Tetragon for runtime protection.
Unable to patch IOT devices.	Segment with Secure Workload / FTD and apply Isovalent/Tetragon runtime protection as compensating controls.
Too many disparate security management interfaces.	Cisco Cloud Control (SCC) unifies multiple UIs into one to simplify policy creation and deployment.

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	EXPECTED CISCO RESPONSE
Limited budget; leverage native features; open to virtual appliances.	Use Cisco Secure Workload to discover and manage workloads and apps; CSW integrates with AWS, Azure and others on access-control policies, and can sync policies between CSW and FTDv virtual appliances.
CIO seeking an end-to-end segmentation solution.	ISE common policy shares context via pxGrid to CSW and FTD — ISE becomes an end-to-end segmentation policy engine.
Minimize complexity.	ISE common policy and CSW discovery unify DC firewalling and policy control.

Reference answer — expected products

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
Cisco Secure Workload (CSW)	Agent and agentless design; App Dependency Mapping (discovery); auto-baseline policy creation; granular host policy enforcement; SGT integration with ISE; AWS/Azure platform integration.	CSW provides comprehensive end-to-end policy.

PRODUCT / FEATURE	HOW IT WORKS	WHY / DIFFERENTIATOR
NGFW / FTD	FTD at the edge for agentless networks; flow info from FTD profiles CSW's traffic baseline; CSW pushes auto-policy to FTD.	Cisco's CSW+FTD integration provides the best end-to-end solution.
Isovalent	eBPF for kernel-level protection; Cilium CNI for Kubernetes (observability, security, LB, BGP); Tetragon for runtime protection.	Kernel-level protection via eBPF is a differentiator.
ISE (common policy)	Shares identity/SGT context via pxGrid to CSW and FTD.	One end-to-end segmentation policy engine.
Cisco Cloud Control (SCC)	Unifies multiple management UIs into one.	Simplifies policy creation and deployment; minimizes complexity.

Zero Trust Access Design Clinic — Solution Guide (confidential). Answers only — pair with the Grading Guide.